A DATA-DRIVEN APPROACH TO ANALYZING GREENHOUSE MICROCLIMATE CONDITIONS USING OPEN DATA SOURCES

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ABSTRACT

Understanding the greenhouse microclimate is essential to gain insight into the performance of controlled environments in agriculture. This study presents a statistical analysis of data collected on the environmental conditions within a greenhouse using **the Greenhouse Shrivan Iran Dataset** (Version 1) from the Mendeley Data repository. The dataset **contained 130,195** observations and 3**7 variables** , and recorded **from 19 December 2023 to 25 May 2024**. The data was analysed using Google Colab in Python to determine quality, descriptive statistics, daily and monthly trends, visualisations and Pearson correlation.

It was found that there was a large difference between the temperature inside and outside the greenhouse. On average the temperature inside the greenhouse was **21.32°C**, whereas it was **8.78°C outside**. This resulted in an average difference of 12.13°C across 150 daily observations. **The** average **relative humidity inside** the greenhouse was **63.90% and 74.21% outside.** The monthly analysis found that the temperature and relative humidity changed over time. **The Pearson correlation** determined that the daily mean readings from **L1** had a **positive relationship** with the **temperature inside** the greenhouse **(r = 0.6955).** There was a negative relationship between the temperature inside the greenhouse and the relative humidity inside **(r= -0.6683).** There was a strong positive relationship between the temperature outside and the temperature **inside (r = 0.8591).**

Overall, this gives a numeric summary of the greenhouse environment and shows how using open access sensor data can be used to analyse the microclimate. Challenges were found when interpreting the data, including no full sensor dictionary being provided and uncertainty around the definition of measurements within the coded variables. This research provides a basis for further analysis surrounding modelling the greenhouse environment, using sensors to monitor and smartly manage a greenhouse using data.

Keywords: Greenhouse microclimate, Environmental monitoring, Temperature variation, Relative humidity, Sensor data analysis, Pearson correlation, Data driven agriculture.

1. Introduction

1.1 Background to study

Greenhouse farming has become an important method for controlling environmental factors affecting growth and productivity. Temperature, relative humidity, light intensity and air flow are important environmental factors.

To create the correct conditions, environmental factors need to be constantly monitored. In some cases, there could be significant differences between the environmental factors inside and outside the greenhouse, meaning that changes in outdoor conditions could relate to changes in the greenhouse. Environmental monitoring can provide insight into researchers about how the greenhouse microclimate behaves.

Recent advances in sensor technologies, IoT, and data analysis have enhanced data collection and analysis capabilities in controlling greenhouse environments. Data from monitoring systems can be collected continuously using sensors (Li *et a*l., 2021, Bhujel *et al*., 2020).

In addition, data analysis can give insight into the trends that occur within the data such as daily and monthly trends. It can also compare conditions inside the greenhouse to outside environments and show statistical relationships between the variables being monitored. This can provide a basis for future modelling, prediction and intelligent control work within greenhouses.

1.2 Problem Statement

Greenhouse environments are affected by multiple environmental factors, and understanding the variation and relationships over time requires analysis of recorded monitoring data. A greenhouse monitoring dataset will contain many observations but only provides useful information if it has been properly prepared, assessed for quality and analyzed. The dataset may contain missing observations, outliers, ambiguous variable definitions and varying coverage of data. A systematic data-driven investigation is required to understand the environmental conditions within a greenhouse. This investigation will help identify trends over time and determine relationships between selected environmental variables. This study will address these issues using a freely available greenhouse monitoring dataset from Shirvan, Iran.

1.3 Aim of the Study

The aim of this study is to analyze greenhouse environmental monitoring data using Python-based statistical and visualization methods. This will be done to investigate trends in temperature, relative humidity and sensor data and identify any correlations between specific variables.

1.4 Objectives of the Study

In this study, the following objectives will be achieved:

i. Analyze and prepare the greenhouse environmental data set.
ii. Evaluate the quality of data by investigating missing or duplicated values and issues with the time stamping.
iii. Investigate trends in temperature inside and outside the greenhouse on a daily basis.
iv. Investigate trends in relative humidity inside and outside the greenhouse on a daily and monthly basis.
v. Calculate the difference between temperatures inside and outside the greenhouse on a daily basis.
vi. Investigate trends in specific sensor data (coded data) such as L1 and WS1.
vii. Investigate any correlations between specific variables through Pearson correlation.
viii. Identify any limitations within the data set and suggest potential areas for future research into monitoring and modelling greenhouses.

1.5 Limitations of the Study

This study will focus on the statistical analysis of a public data set of greenhouse environmental monitoring data collected in Shirvan, Northern Khorasan, Iran. The data set will be analyzed for the period of 19th December 2023 to 25th May 2024.

The analysis will focus on temperature inside and outside the greenhouse, relative humidity inside and outside the greenhouse and specific coded sensor data. This will be done by calculating the average temperature and humidity inside and outside on a daily basis, calculating descriptive

statistics and visualizing the data. Pearson correlation will also be applied to identify any correlations between environmental variables. This study is limited to the data provided. No new experiments will be conducted in the greenhouse. There will be no calibration of sensors. Crop yield will not be assessed. A validated control system will not be developed. The exact definition and units of some variables could not be verified so these will be considered as coded sensor data.

1.6 Significance of the Study

This study shows how public data sets can be utilized to investigate environmental variables within the greenhouse through data analysis. An overview of trends in temperature and humidity has been provided and any correlations between specific variables have been identified.

This process could be used as a basis for further research into modelling environmental variables in a greenhouse, analyzing sensor data, implementing smart monitoring systems and managing the environment through data.

2. Literature Review

The greenhouse microclimate can be affected by many factors. These include air temperature, relative humidity, light levels, and air flow.

Boulard, T. & Wang, Y. (2002) examined how greenhouse climate models have developed and the importance of understanding heat and mass transfer. Temperature, humidity levels, ventilation, etc. were considered within this research. This provides a basis for the mathematical analysis of greenhouse conditions.

Shamshiri *et al.* (2018) investigated recent advancements in automation within greenhouses. Sensors, automation and environmental monitoring were explored in this study. It was shown that maintaining the correct environmental conditions will increase production efficiency.

Bhujel *et al.* (2020) examined sensor systems within greenhouses for monitoring and controlling microclimates. Different types of sensors, placement of sensors and monitoring methods for environmental conditions such as temperature, humidity and solar radiation were discussed throughout the research. They highlight the importance of choosing the correct type of sensor in the right position for accurate monitoring of environmental conditions to be achieved.

Ouammi *et al.* (2020) investigated a decision support system for optimizing microclimate monitoring within smart connected greenhouses. Correct sensor placement was looked into to achieve efficient monitoring of environmental conditions. This shows the importance of having sensors placed correctly.

Shamshiri *et al.* (2020) investigated the modelling of greenhouse microclimates using IoT sensor data fusion. A comfort ratio model was developed to help understand the conditions within a greenhouse. This can be used to aid efficient crop production using less energy. They show how combining sensor data with mathematical models can assist with understanding environmental conditions.

Li *et al.* (2021) examined methods for monitoring greenhouses using IoT technologies. Various ways of monitoring environmental conditions within greenhouses were discussed. Sensor technologies were looked at along with ways of measuring environmental conditions, soil

conditions and crop conditions. Automated monitoring systems were developed. Key areas for research have been identified.

Badji *et al.* (2022) investigated the design, technologies and management of greenhouses. Various aspects were looked at including design, technology development and management. Correct technologies and management methods are important to improve performance.

Brentarolli *et al.* (2024) examined a spatiotemporal approach for mapping microclimates within greenhouses. Non-uniform environmental conditions within greenhouses were investigated. This showed that spatiotemporal differences must be considered during microclimate analysis.

Akintan *et al.* (2025) introduced a digital twin approach for sensor selection and microclimate monitoring. Distributed temperature and relative humidity sensors were used to look at adaptive sensor selection. This demonstrates how digital twins can reduce redundant sensors.

Bicamumakuba *et al.* (2025) examined multi-sensor monitoring, intelligent control and data processing methods for managing smart greenhouses. Environmental sensors, automation using IoT, artificial intelligence and data filtering were looked at. Sensor integration was found to be important along with accurate data processing. Calibration issues, interoperability and computational issues remain issues.

Yu *et al.* (2025) looked into methods for predicting and controlling temperature within greenhouses. Various sensor technologies, data fusion, traditional modelling and artificial intelligence were explored. Predictive models and intelligent control systems can be used to manage temperature.

2.1 Research Gap

Many studies have looked into sensor technologies within greenhouses, automation, modelling microclimates and intelligent control. More work is required to demonstrate how publicly available greenhouse datasets can be analyzed using replicable statistical methods. Investigating the relationship between internal and external temperatures, relative humidity, daily and monthly variations and the relationship between sensor variables would be a useful area to explore. Issues can arise when trying to interpret sensor variables due to incomplete descriptions of the coded variables. This study looks at the Greenhouse Shrivan Iran Dataset, using Python for data

processing, quality checks, descriptive statistics, time series visualization and Pearson correlation. Limitations to interpreting the sensor variables are identified.

3. Materials and Methods

3.1 Study Area and Dataset

This dataset has been sourced from the publicly available Greenhouse Shrivan Iran Dataset from Mendeley Data (DOI: 10.17632/ncrm9p4tc8.1). The data was collected from a greenhouse located in Shirvan, Northern Khorasan, Iran. The data includes environmental data such as inside and outside temperature, inside and outside relative humidity, light, wind, and coded variables. There **were 130,195 rows and 37 columns** within the dataset. The timestamps ranged from 19 December 2023 to 25 May 2024. The dataset description states that it was collected between November 2023 to April 2024 however this dataset has timestamps up until May 2024.

3.2 Data Loading

The CSV dataset was uploaded into a Python environment using Google colab. The data was analysed to understand the dataset including the size, column names, variable types, missing data, duplicated and valid timestamps. The DateTime column was converted to a datetime variable. The data was analysed to see if there were any duplicated rows, duplicated timestamps, and missing data. The Unnamed: 0 column was identified as an index column and not an environmental variable.

3.3 Environmental Variables

The focus of this analysis was to understand the following environmental variables:

i. MeanInsideTemp – mean inside temperature (°C).
ii. OutsideTemp – outside temperature (°C).
iii. AVGHumidityInside – average inside relative humidity (%).
iv. HumidityOutside – outside relative humidity (%).
v. L1 – coded sensor variable, which may relate to light intensity.
vi. WS1 – coded sensor variable, which may relate to wind speed.

The description does state that light intensity and wind speed were monitored, however the exact meaning and units for L1 and WS1 could not be determined from a variable level data dictionary. Therefore, these variables have been treated as coded sensor variables.

Variables F, H and W have not been given a clear meaning to what equipment or control system they relate to, due to the lack of understanding of what the individual variable means.

3.4 Data Quality

Missing data, duplicated rows, duplicated timestamps, and invalid datetime variables were checked. There were no duplicated rows, duplicated timestamps or invalid timestamps. There was however a small amount of missing data across multiple sensor variables. The L1 variable had some odd readings with 0 and high readings. These were kept in the dataset for initial analysis since we could not confirm what they meant or how they were measured/calibrated.

3.5 Descriptive Statistics

Basic descriptive statistics were carried out on the data. This included mean, standard deviation, min, max and specific percentiles. Daily averages were calculated to understand how the conditions within the greenhouse varied across different days. Monthly averages were also calculated to compare how the inside and outside temperatures and relative humidity varied across different months. The difference between the inside and outside temperature on a daily basis by subtracting the outside temperature from the inside temperature.

3.6 Correlation

Pearson correlation was used to understand the linear relationships between select variables. This included:

i. Daily mean L1 and daily mean inside temperature.
ii. Daily mean inside temperature and daily mean inside relative humidity.
iii. Daily mean outside temperature and daily mean inside temperature.

Correlation coefficients and p-values were calculated. This shows the relationship between the variables however this is not causal, just a statistical relationship. As this is a time series the data may have some autocorrelation.

3.7 Software Used

Python was used in Google colab to process, analyze and visualize the data. The analysis was completed using Python tools to manipulate the data, calculate numerical, statistical and plot graphs to show trends across days, comparisons across months and relationships between selected variables.

3.8 Limitations

The lack of full data dictionary for the sensor variables limited the analysis. Several of the coded sensor variables could not be confirmed what they meant or what units they were in. The timestamps did not match the description stating the data was collected between November 2023 to April 2024.

The data only relates to one location within a greenhouse and was only collected over a short period of time. This analysis should therefore be seen to relate to the data provided and not generalize to all greenhouses.

4. Results and Discussion

4.1 Greenhouse Dataset

The dataset that was downloaded included 130,195 rows of data and 37 columns. The times recorded range between 19 December 2023 and 25 May 2024. The dataset contains information on environmental factors within the greenhouse (internal/external temperature, relative humidity, amongst others). There are several coded sensor readings.

Overall, there were no issues with duplicate rows, duplicate times or invalid date time data. However, there were a few missing values in some of the sensors. These have been noted but not imputed.

Table 4.1. Summary of the Greenhouse Dataset

Parameter	Description
Dataset name	Greenhouse Shrivan Iran Dataset
Total records	130,195
Total columns	37
Start date	19 December 2023
End date	25 May 2024
Duplicate rows	0
Duplicate timestamps	0
Invalid timestamps	0
Data format	CSV

4.2 Internal and External Temperature

From the analysis there were clear differences between the internal and external temperatures. The average temperature inside was 21.32 °C and the average temperature outside was 8.78 °C. The internal temperature ranged between 5.5 °C and 41.0 °C. Whereas the external temperature ranged between -9.4 °C and 37.3 °C.

This shows that throughout the time the greenhouse had a higher temperature inside compared to the outside. Looking at the temperature trends throughout each day there was a much more consistent temperature inside the greenhouse compared to outside.

Table 4.2. Descriptive Statistics of Inside and Outside Temperature

Statistical parameter	Inside temperature (°C)	Outside temperature (°C)
Number of observations	130,194	130,195
Mean	21.32	8.78
Standard deviation	3.72	7.97
Minimum	5.5	−9.4
5th percentile	17.4	−2.0
25th percentile	19	2.4
Median	20.3	7.8
75th percentile	22.7	14.1
95th percentile	29.5	23.8
Maximum	41	37.3

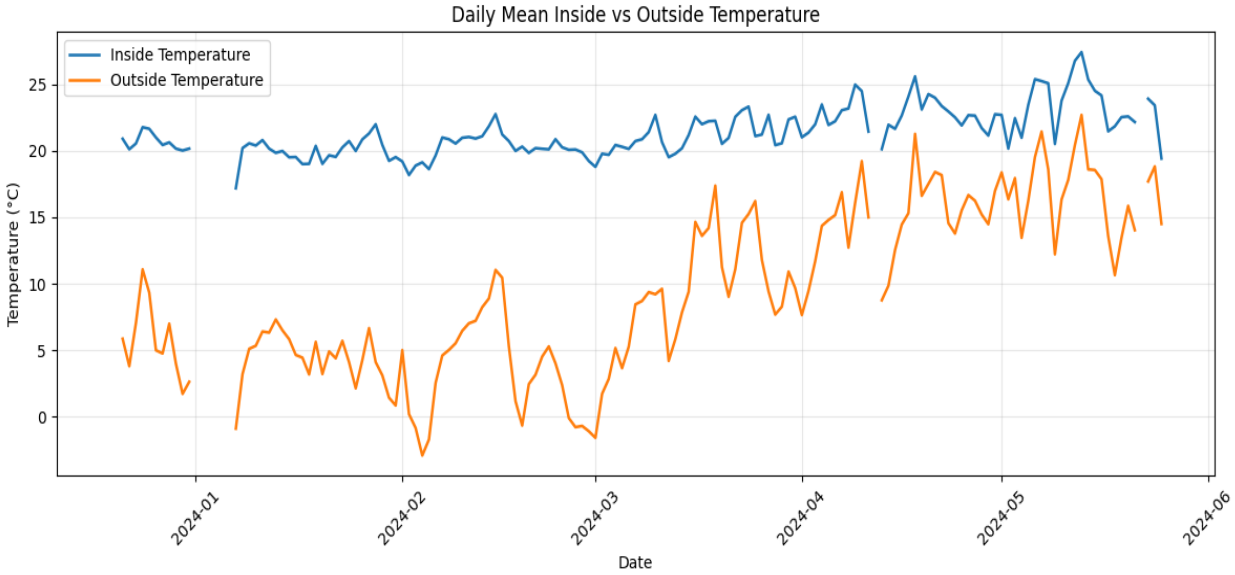


Figure 4.1. Daily Mean Inside and Outside Temperature

The temperature difference between the outside and the inside of the building per day was calculated by subtracting the daily average outdoor temperature from the daily average indoor temperature. On average the temperature difference was 12.13°C across the 150 days of data. There was a minimum temperature difference of 3.80°C and maximum temperature difference of 22.06°C. All the days had a greater temperature inside the building compared to outside.

Table 4.3. Statistical Summary of Daily Inside–Outside Temperature Difference

Parameter	Temperature difference (°C)
Number of daily observations	150
Mean	12.13
Minimum	3.8
Maximum	22.06
Days with positive difference	150
Days with zero or negative difference	0

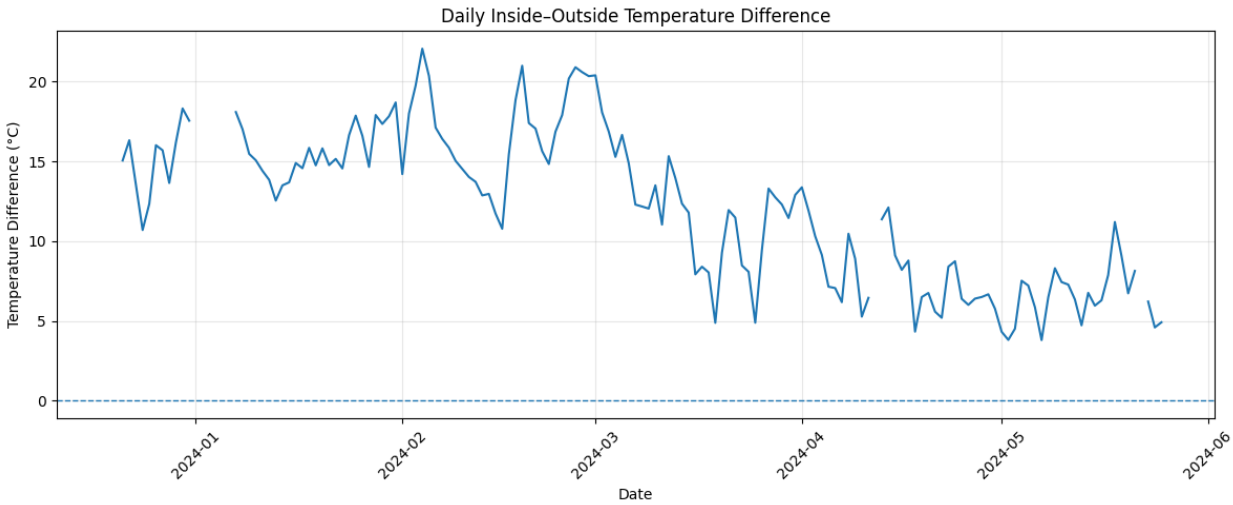


Figure 4.2. Daily Mean Inside–Outside Temperature Difference

4.3 Relative Humidity

The average relative humidity levels inside were 63.90% and outside were 74.21%. This shows a larger variation of data outside with a greater standard deviation of 25.36 than inside with a standard deviation of 18.98. From looking at the data each day it can be seen that there were more days (119) when the relative humidity was higher outside than inside (31). This shows that overall, the outside environment was more humid than inside, however it does vary day to day.

Table 4.4. Descriptive Statistics of Inside and Outside Relative Humidity

Statistical parameter	Inside relative humidity (%)	Outside relative humidity (%)
Number of observations	130,195	130,194
Mean	63.9	74.21
Standard deviation	18.98	25.36
Minimum	1.1	0
5th percentile	27.3	28

25th percentile	51.9	54.7
Median	67.2	80.6
75th percentile	79.2	100
95th percentile	88.7	100
Maximum	100	100

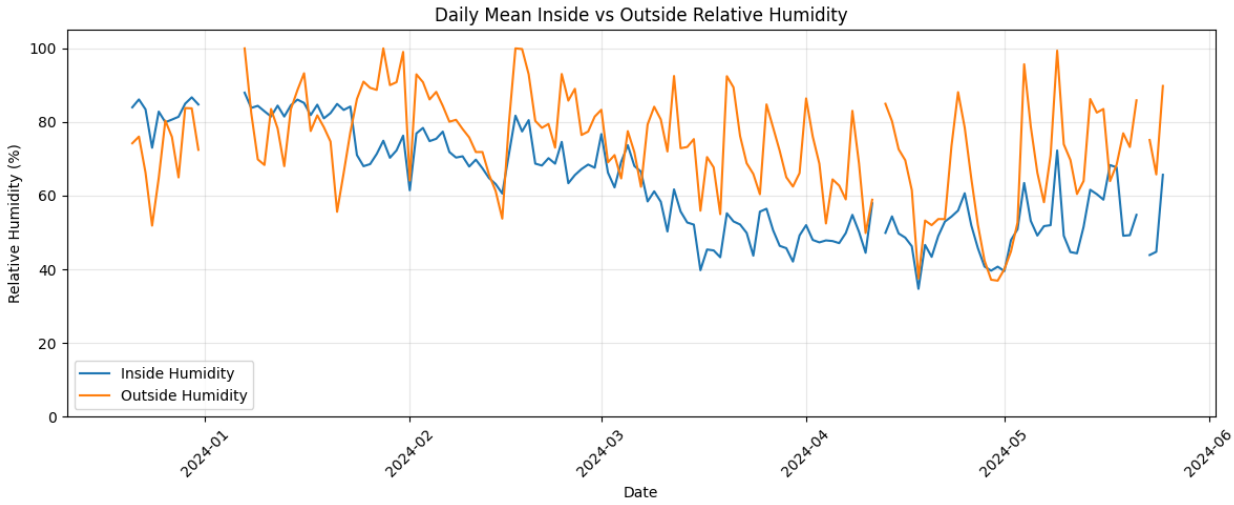


Figure 4.3. Daily Mean Inside and Outside Relative Humidity

4.4 Relationship Between Light Sensor Values and Internal Temperature

A positive relationship was found between the mean daily values of the L1 sensor and the mean daily internal temperature. Using Pearson's correlation, a strong positive linear relationship was found between the two variables. This gives a correlation coefficient of r=0.6955 with a p-value of approx. 5.25×10^{-23}, this was calculated using 150 observations.

However, this only shows that there is a moderate to strong positive relationship between the two variables in this data set. We cannot confirm that the L1 variable is giving us a measurement of light intensity as we do not know what units it is measuring in and how it maps to the sensor.

Additionally, we cannot confirm that changes in the L1 variable caused changes in greenhouse temperature.

Table 4.5. Pearson Correlation Between Daily Mean L1 Reading and Inside Temperature

Statistical parameter	Result
Number of daily observations	150
Pearson correlation coefficient (r)	0.6955
p-value	5.25×10^{-23}
Relationship	Positive association

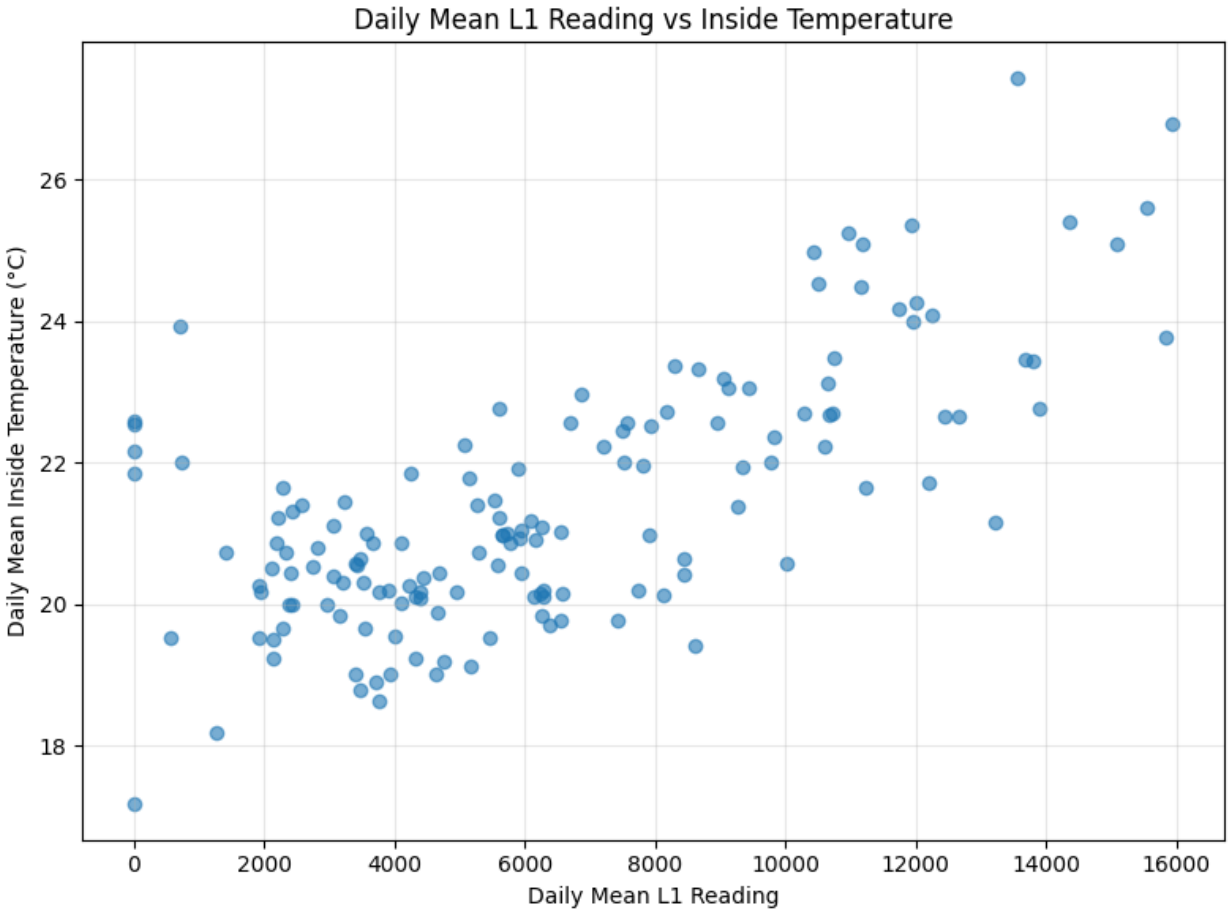


Figure 4.4: Relationship Between Daily Mean L1 Reading and Inside Temperature

4.5 Correlation between internal temperature and internal relative humidity

There appears to be a negative correlation between the daily average internal temperature and daily average internal relative humidity. The Pearson correlation analysis shows there is a correlation coefficient of -0.6683, p-value of approximately 9.31×10^{-21}, based on 150 daily observations.

This means the daily average internal temperature tends to increase when the daily average internal relative humidity decreases. This does not show a definite cause and effect relationship.

Table 4.6. Pearson Correlation Between Inside Temperature and Inside Relative Humidity

Statistical parameter	Result
Number of daily observations	150
Pearson correlation coefficient (r)	−0.6683
p-value	9.31×10^{-2}
Relationship	Negative association

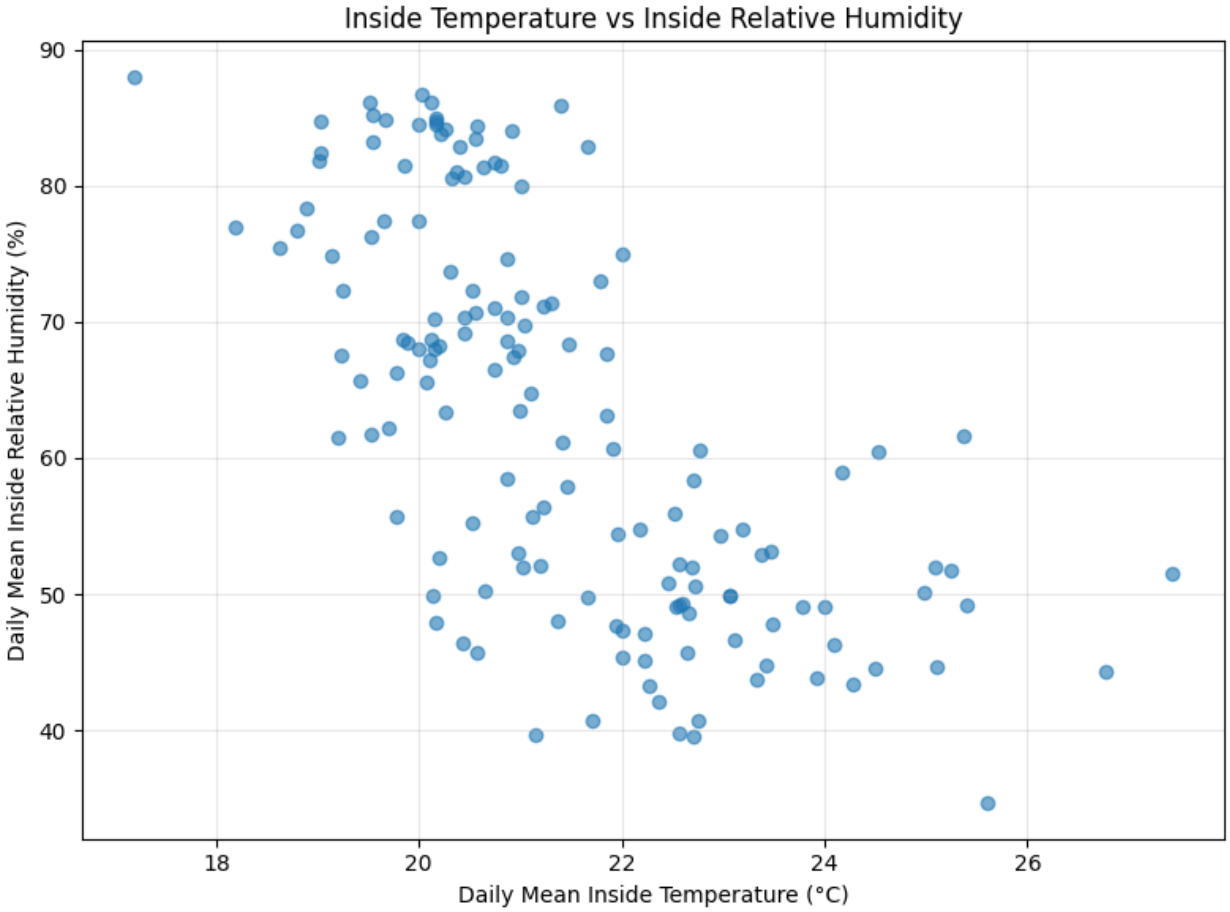


Figure 4.5. Relationship Between Daily Mean Inside Temperature and Inside Relative Humidity

4.6 Relationship Between Outside and Inside Temperature

The daily mean outside temperature had a strong positive relationship with the daily mean inside temperature. This can be seen by Pearson's correlation coefficient (0.8591) with a p-value approximately 6.78×10^{-45}, of based on 150 observations. This suggests that as the outside temperature increased so did the inside temperature. It shows that there is a clear link between the outside environments and greenhouse temperatures. However, this only shows correlation and not causation. It does not show which of the greenhouse control systems contributed to this.

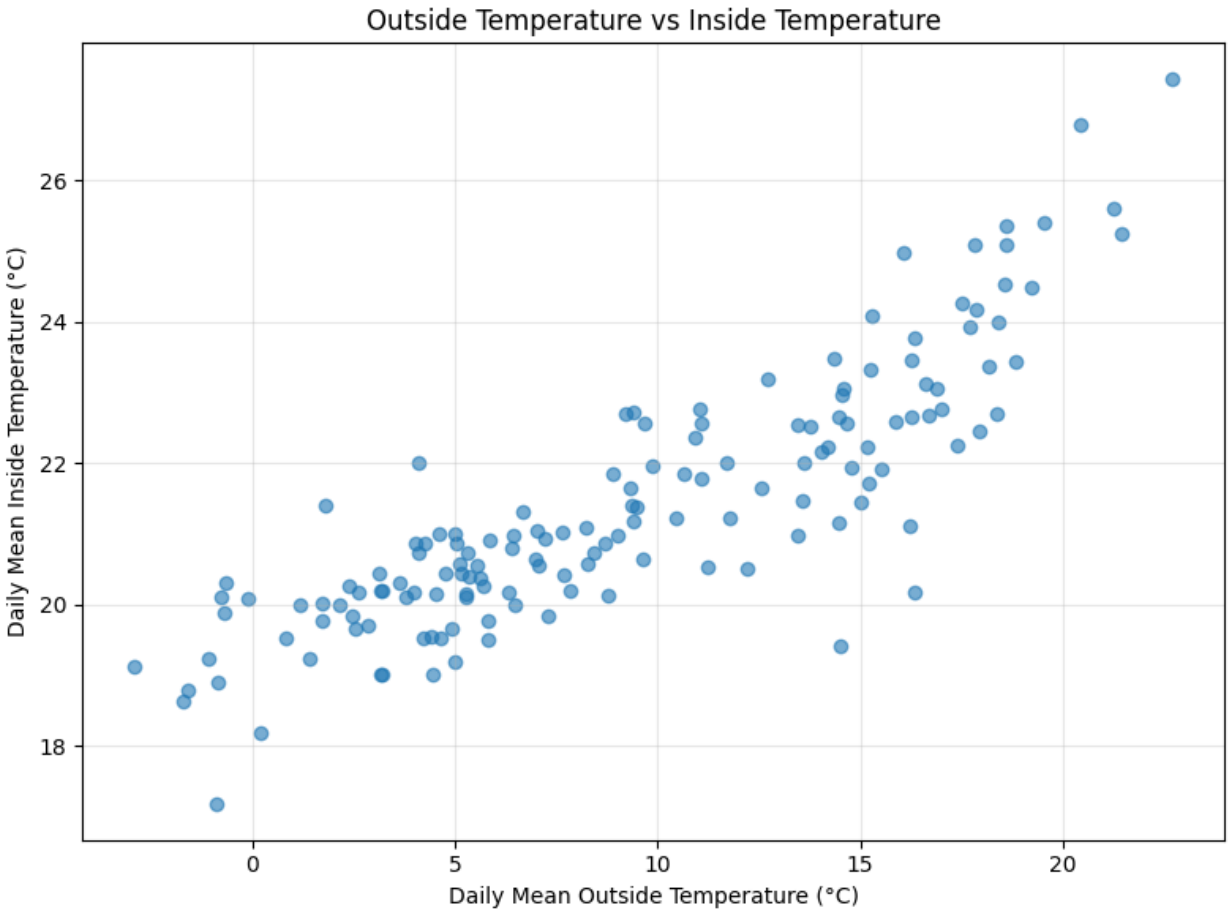


Figure 4.6. Relationship Between Daily Mean Outside and Inside Temperature

4.7 Time Series Analysis of WS1 Sensor Reading

There were 130,194 valid observations recorded for the variable WS1, with a mean value of 7.19 and standard deviation of 7.00. The variable ranged between 0 and 87. The mean value for each day was calculated and there were 150 days recorded. This had a mean value of 7.24, a minimum of 2.25 and a maximum of 20.76.

The data from the repository indicates that one of the environmental factors being monitored is wind speed. However, it was not possible to independently verify what the WS1 variable corresponded to or what units were used. As such the results are presented as readings from the WS1 sensor.

Table 4.7. Statistical Summary of the WS1 Sensor Reading

Statistical parameter	Result
Valid observations	130,194
Raw mean	7.19
Raw standard deviation	7
Raw minimum	0
Raw maximum	87
Daily observations	150
Daily mean average	7.24
Minimum daily mean	2.25
Maximum daily mean	20.76

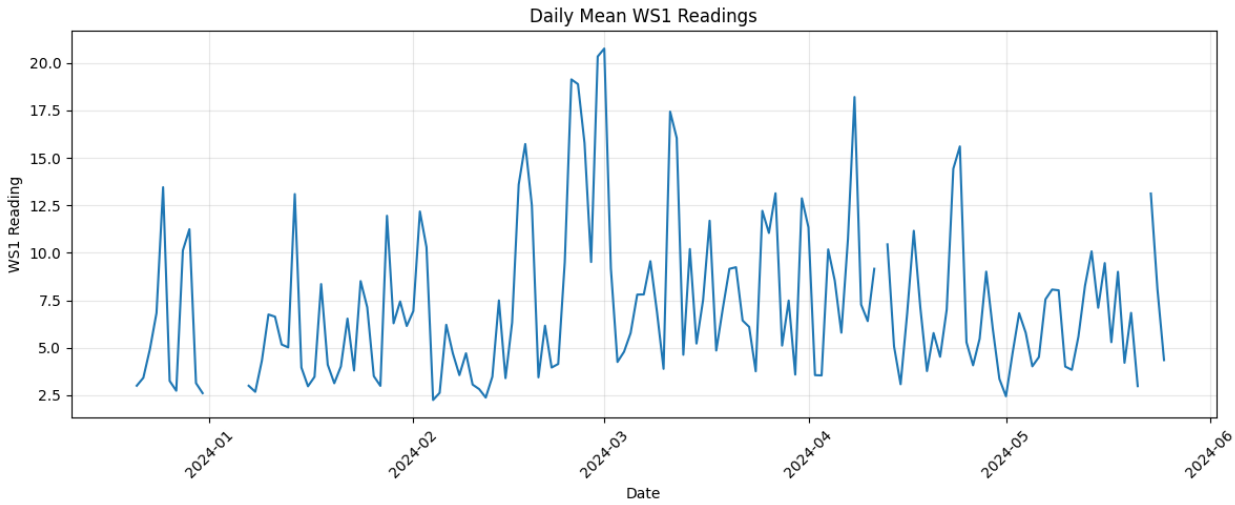


Figure 4.7. Daily Mean WS1 Sensor Readings

4.8 Monthly Variation in Greenhouse Temperature

The data shows that both the temperature inside and outside the greenhouse have been increasing gradually over the months. The average temperature inside increased from 20.70°C in December 2023 to 23.53°C in May 2024. The average temperature outside increased from 5.37°C to 16.98°C.

The temperature inside was always higher than the temperature outside throughout the months recorded. This agrees with the previous analysis of the temperature each day. This shows that the temperature inside the greenhouse has been warmer than outside.

Table 4.8. Monthly Average Inside and Outside Temperature

Month	Inside temperature (°C)	Outside temperature (°C)
December 2023	20.7	5.37
January 2024	20.09	4.47
February 2024	20.3	3.52
March 2024	21.12	8.92
April 2024	22.71	14.86
May 2024	23.53	16.98

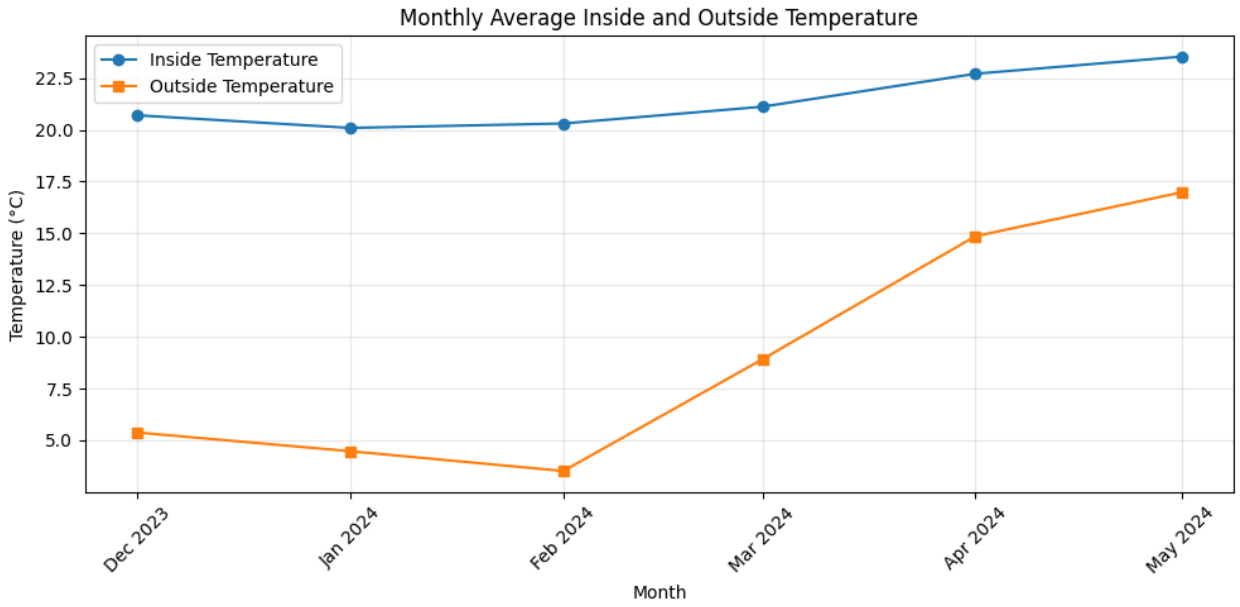


Figure 4.8. Monthly Average Inside and Outside Temperature

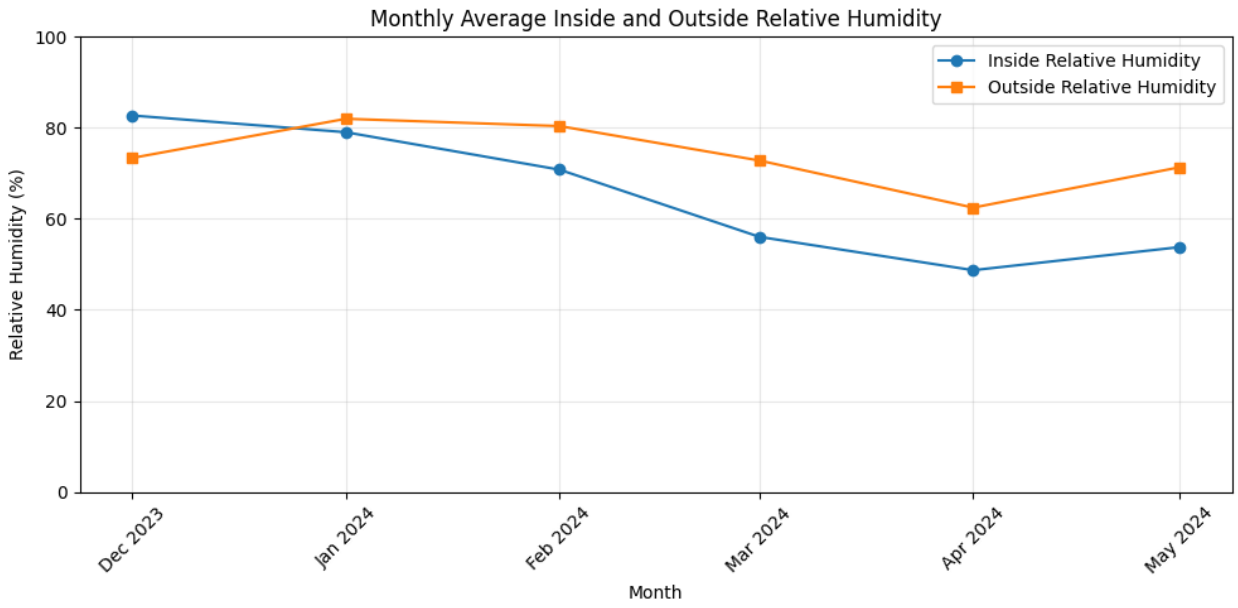
4.9 Monthly Fluctuations of Relative Humidity

The mean relative humidity levels inside decreased between December 2023 (82.72%) and April 2024 (48.72%) but then increased again in May 2024 (53.80%). Whereas the relative humidity levels outside decreased between January 2024 (81.99%) and April 2024 (62.47%), however increased in May 2024 (71.36%).

This shows that throughout the time humidity levels have fluctuated. The trend of the relative humidity inside did not match exactly with the trend outside.

Table 4.9. Monthly Average Inside and Outside Relative Humidity

Month	Inside relative humidity (%)	Outside relative humidity (%)
December 2023	82.72	73.38
January 2024	79.04	81.99
February 2024	70.81	80.39
March 2024	56.04	72.82
April 2024	48.72	62.47
May 2024	53.8	71.36



4.10 Data Quality Issues and Interpretation

The data within the L1 data include many zeroes. There are also several extreme values with values of 65,535. This means that the average values per day are also very low/zero in parts of May. As there is no information about how the sensors have been calibrated, or detailed variable data dictionary it is not possible to determine if the data represents real environmental conditions, sensor restrictions, missing data codes or measurement issues.

The precise meaning of the coded F, H and W columns could not be determined so they were removed from the equipment specific interpretation.

The correlation analysis has been conducted using daily averages of the data. Environmental data can often be autocorrelated over time, so the p-values should be considered as initial statistical evidence. Further analysis using time series techniques and validated sensor metadata would need to be done to make stronger conclusions.

Overall, the results provide a description of the greenhouse environmental trends and statistical relationships within the available data. This will help provide a foundation for looking further into how to monitor and control the environmental conditions within greenhouses. However, the lack of sensor documentation will need to be considered.

5. Conclusion and Future Recommendations

5.1 Conclusion

In this study the Greenhouse Shrivan Iran Dataset was explored to understand the environmental conditions within greenhouses and correlations between specific environmental variables. This dataset had 130,195 entries between dates 19 December 2023 and 25 May 2024. Data was analysed through statistics, daily trends, monthly comparisons and correlation analysis using Python in Google Colab. It was found that on average the temperature inside the greenhouse was at 21.32°C and outside at 8.78°C. The internal temperatures were always higher than the external temperatures across all 150 days in the daily comparison. There was an average temperature difference of 12.13°C between the outside and inside of the greenhouse.

For humidity levels the average internal humidity was 63.90% and external humidity was 74.21%. The external humidity was always higher on 119 days however the internal was higher on 31 days. Monthly analysis showed temperature and humidity varied throughout.

From the correlation analysis it was found that there was a positive correlation between the daily mean of L1 and internal temperature ($r = 0.6955$). There was a negative correlation between internal temperature and internal relative humidity ($r=-0.6683$). There was a strong positive correlation between external and internal temperature ($r = 0.8591$). However this does not represent cause and affect just correlations between variables.

Overall, this study has provided an insight into the environmental conditions within greenhouses using public data. It has shown how Python can be used for data processing and visualisation to understand temperature, humidity and sensor patterns within greenhouses.

5.2 Limitations

Limitations of this research include:

There was no full data dictionary provided for the variables therefore it was not possible to confirm the actual meanings and units for L1, WS1, F, H and W variables.

The time period stated in the repository description did not match the time period in the CSV file. The time period in the repository description was November 2023 to April 2024 whereas the data in the CSV file was between 19 December 2023 and 25 May 2024.

Within the L1 variable there were many zeroes and some high values. Without knowing the calibration of the sensors it is unclear how they would translate to real values.

The correlation analysis was based on the daily mean of 150 observations. This could be affected by auto-correlation. Further time series analysis is required.

The data only represents one set of greenhouses and a small amount of observations meaning results cannot be generalised to all greenhouses.

5.3 Recommendations for Future Research

Further research could be done to expand on this study by:

1. Sensor information and calibration. Obtain full data dictionary for sensors to confirm units, calibration and what each coded variable means in operation.

2. Multi-year analysis. Analyse data over multiple years to see how seasonal and yearly changes impact greenhouse environmental conditions.

3. Time series analysis. Perform correct time series analysis to understand patterns, auto-correlation and correlations between variables.

4. Modelling greenhouse environmental conditions. Create models to understand how external conditions relate to internal temperature and humidity.

5. Smart greenhouse control. Understand how the verified environmental conditions could be used to monitor and make decisions about heating, ventilation, windows etc within greenhouses.

6. Renewable energy integration. Investigate renewable energy technologies that could be integrated into greenhouses.

This could provide a better understanding of how the environmental conditions behave within greenhouses and provide further research into smart and energy efficient greenhouses.

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